

## **Modbus table for connecting S21 automation to BMS**

**TO ENABLE OPERATION VIA THE MODBUS RTU PROTOCOL THROUGH THE RS-485 INTERFACE, DISCONNECT ALL THE WIRED CONTROL PANELS CONNECTED TO THE AIR HANDLING UNIT THROUGH THIS INTERFACE**

**SIMULTANEOUS OPERATION THROUGH RS-485, WI-FI, AND ETHERNET INTERFACES IS POSSIBLE**

**TO USE WIRED CONTROL PANELS, THE BMS MUST BE CONNECTED THROUGH WI-FI AND/OR ETHERNET INTERFACES VIA MODBUS TCP PROTOCOL**

#### MODBUS PARAMETERS

| Modbus RTU          |                     |                                                |                        |                |
|---------------------|---------------------|------------------------------------------------|------------------------|----------------|
| Baud rate           | Number of data bits | Stop bits                                      | Parity type            | Address        |
| 9600                | 8                   | 1                                              | None (by default)      | 1-16           |
| 14400               |                     | 1.5                                            | even                   | 1 (by default) |
| 19200               |                     | 2 (by default)                                 | odd                    |                |
| 38400               |                     |                                                |                        |                |
| 57600               |                     |                                                |                        |                |
| 115200 (by default) |                     |                                                |                        |                |
| Modbus TCP          |                     |                                                |                        |                |
| IP address*         | Port                | Maximum number of simultaneous TCP connections | TCP connection timeout |                |
| Static              | 502                 | For Ethernet = 1, for Wi-Fi = 1                | 30 seconds             |                |
| DHCP (by default)   |                     |                                                |                        |                |

\*Wi-Fi IP address in access point mode – 192.168.4.1

The RS-485, Wi-Fi, and Ethernet network parameters for air handling units are configured using a mobile application.  
 Maximum number of registers in one package: 125 (for 16-bit registers) and 2000 (for 1-bit registers).  
 Supported modbus functions: 1, 2, 3, 4, 5, 6, 15, 16.

| Address                                                        | R/W | Variable                   | Description                                                                     | Minimum value | Maximum value | Pre-set value | Measurement units | Model | Dimension |
|----------------------------------------------------------------|-----|----------------------------|---------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|-------|-----------|
| <b>Coils (1 bit registers) - modbus functions: 1, 5, 15</b>    |     |                            |                                                                                 |               |               |               |                   |       |           |
| 0                                                              | R/W | CL_POWER                   | Coils (1-bit registers) - modbus functions: 1, 5, 15                            | 0             | 1             | 0             | —                 | Bool  | 1         |
| 1                                                              | R/W | CL_TIMER                   | Unit On/Off                                                                     | 0             | 1             | 0             | —                 | Bool  | 1         |
| 2                                                              | R/W | CL_WEEK                    | Main timer                                                                      | 0             | 1             | 0             | —                 | Bool  | 1         |
| 3                                                              | R   | CL_Boost_MODE              | Weekly Schedule                                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 4                                                              | R   | CL_FPLC_MODE               | Boost mode                                                                      | 0             | 1             | —             | —                 | Bool  | 1         |
| 5                                                              | R/W | CL_IntRH_CTRL              | Fireplace mode                                                                  | 0             | 1             | 0             | —                 | Bool  | 1         |
| 6                                                              | R/W | CL_ExtRH_CTRL              | Main humidity sensor activation                                                 | 0             | 1             | 0             | —                 | Bool  | 1         |
| 7                                                              | R/W | CL_IntCO2_CTRL             | External humidity sensor activation                                             | 0             | 1             | 0             | —                 | Bool  | 1         |
| 8                                                              | R/W | CL_ExtCO2_CTRL             | Main CO <sub>2</sub> sensor activation                                          | 0             | 1             | 0             | —                 | Bool  | 1         |
| 9                                                              | R/W | CL_IntPM2_5_CTRL           | External PM2.5 sensor activation                                                | 0             | 1             | 0             | —                 | Bool  | 1         |
| 10                                                             | R/W | CL_ExtPM2_5_CTRL           | Main PM2.5 sensor activation                                                    | 0             | 1             | 0             | —                 | Bool  | 1         |
| 11                                                             | R/W | CL_IntVOC_CTRL             | External PM2.5 sensor activation                                                | 0             | 1             | 0             | —                 | Bool  | 1         |
| 12                                                             | R/W | CL_ExtVOC_CTRL             | Main VOC sensor activation                                                      | 0             | 1             | 0             | —                 | Bool  | 1         |
| 13                                                             | R/W | CL_BoostSWITCH_CTRL        | External VOC sensor activation                                                  | 0             | 1             | 1             | —                 | Bool  | 1         |
| 14                                                             | R/W | CL_FplcSWITCH_CTRL         | Input activation for the boost mode switch                                      | 0             | 1             | 1             | —                 | Bool  | 1         |
| 15                                                             | R/W | CL_FireALARM_CTRL          | Input activation for the fireplace mode switch                                  | 0             | 1             | 0             | —                 | Bool  | 1         |
| 16                                                             | R/W | CL_10V_SENSOR_CTRL         | Fire alarm sensor activation                                                    | 0             | 1             | 0             | —                 | Bool  | 1         |
| 17                                                             | W   | CL_RESET_FILTER_TIMER      | Input activation for the external control device 0-10 V                         | 1             | 1             | —             | —                 | Bool  | 1         |
| 18                                                             | W   | CL_RESET_ALARM             | Reset timer countdown to filter replacement                                     | 1             | 1             | —             | —                 | Bool  | 1         |
| 19                                                             | W   | CL_RESTORE_FACTORY         | Reset all alarms                                                                | 1             | 1             | —             | —                 | Bool  | 1         |
| 20                                                             | R/W | CL_CLOUD_CTRL              | Restore everything to factory settings                                          | 0             | 1             | 0             | —                 | Bool  | 1         |
| 21                                                             | R/W | CL_MinSuAirOutTEMP_CTRL    | Activation of control via cloud server                                          | 0             | 1             | 1             | —                 | Bool  | 1         |
| 22                                                             | R/W | CL_WaterPRESS_CTRL         | Minimum room supply air temperature control                                     | 0             | 1             | 1             | —                 | Bool  | 1         |
| 23                                                             | R/W | CL_WaterFLOW_CTRL          | Heat medium water pressure sensor activation                                    | 0             | 1             | 0             | —                 | Bool  | 1         |
| 24                                                             | R/W | CL_WaterHeaterAutoRestart  | Heat medium water flow sensor activation                                        | 0             | 1             | 1             | —                 | Bool  | 1         |
| <b>Discrete Inputs (1-bit registers) - modbus functions: 2</b> |     |                            |                                                                                 |               |               |               |                   |       |           |
| 0                                                              | R   | DI_CurBoostSWITCH          | Current input status for the Boost mode switch                                  | 0             | 1             | —             | —                 | Bool  | 1         |
| 1                                                              | R   | DI_CurFplcSWITCH           | Current input status for the Fireplace mode switch                              | 0             | 1             | —             | —                 | Bool  | 1         |
| 2                                                              | R   | DI_CurFireALARM            | Current status of the fire alarm sensor                                         | 0             | 1             | —             | —                 | Bool  | 1         |
| 3                                                              | R   | DI_StatusRH                | Humidity setpoint excess indication                                             | 0             | 1             | —             | —                 | Bool  | 1         |
| 4                                                              | R   | DI_StatusCO2               | CO <sub>2</sub> setpoint excess indication                                      | 0             | 1             | —             | —                 | Bool  | 1         |
| 5                                                              | R   | DI_StatusPM2_5             | PM2.5 setpoint excess indication                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 6                                                              | R   | DI_StatusVOC               | VOC setpoint excess indication                                                  | 0             | 1             | —             | —                 | Bool  | 1         |
| 7                                                              | R   | DI_StatusHEATER            | Heater operation indication                                                     | 0             | 1             | —             | —                 | Bool  | 1         |
| 8                                                              | R   | DI_StatusCOOLER            | Cooler operation indication                                                     | 0             | 1             | —             | —                 | Bool  | 1         |
| 9                                                              | R   | DI_StatusFanBLOWING        | Electric heater blowdown indication                                             | 0             | 1             | —             | —                 | Bool  | 1         |
| 10                                                             | R   | DI_CurPreHeaterThermostat  | Current input status for the preheating thermostat                              | 0             | 1             | —             | —                 | Bool  | 1         |
| 11                                                             | R   | DI_CurMainHeaterThermostat | Current input status for the reheating thermostat                               | 0             | 1             | —             | —                 | Bool  | 1         |
| 12                                                             | R   | DI_CurSuFilterPRESS        | Current input status for the differential pressure switch of the supply filter  | 0             | 1             | —             | —                 | Bool  | 1         |
| 13                                                             | R   | DI_CurExFilterPRESS        | Current input status for the differential pressure switch of the extract filter | 0             | 1             | —             | —                 | Bool  | 1         |
| 14                                                             | R   | DI_CurWaterPRESS           | Current status of the heat medium water pressure sensor                         | 0             | 1             | —             | —                 | Bool  | 1         |
| 15                                                             | R   | DI_CurWaterFLOW            | Current status of the heat medium water flow sensor                             | 0             | 1             | —             | —                 | Bool  | 1         |
| 16                                                             | R   | DI_CurSuFanPRESS           | Current input status for the differential pressure switch of the supply fan     | 0             | 1             | —             | —                 | Bool  | 1         |
| 17                                                             | R   | DI_CurExFanPRESS           | Current input status for the differential pressure switch of the extract fan    | 0             | 1             | —             | —                 | Bool  | 1         |
| 18                                                             | R   | DI_WaterPreheatingStatus   | Return water heating indicator before the air handling unit start-up            | 0             | 1             | —             | —                 | Bool  | 1         |
| 19                                                             | R   | DI_AlarmCODE0              | Alarm indicator with code No. 0                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 20                                                             | R   | DI_AlarmCODE1              | Alarm indicator with code No. 1                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 21                                                             | R   | DI_AlarmCODE2              | Alarm indicator with code No. 2                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 22                                                             | R   | DI_AlarmCODE3              | Alarm indicator with code No. 3                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 23                                                             | R   | DI_AlarmCODE4              | Alarm indicator with code No. 4                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 24                                                             | R   | DI_AlarmCODE5              | Alarm indicator with code No. 5                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 25                                                             | R   | DI_AlarmCODE6              | Alarm indicator with code No. 6                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 26                                                             | R   | DI_AlarmCODE7              | Alarm indicator with code No. 7                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 27                                                             | R   | DI_AlarmCODE8              | Alarm indicator with code No. 8                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 28                                                             | R   | DI_AlarmCODE9              | Alarm indicator with code No. 9                                                 | 0             | 1             | —             | —                 | Bool  | 1         |
| 29                                                             | R   | DI_AlarmCODE10             | Alarm indicator with code No. 10                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 30                                                             | R   | DI_AlarmCODE11             | Alarm indicator with code No. 11                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 31                                                             | R   | DI_AlarmCODE12             | Alarm indicator with code No. 12                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 32                                                             | R   | DI_AlarmCODE13             | Alarm indicator with code No. 13                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 33                                                             | R   | DI_AlarmCODE14             | Alarm indicator with code No. 14                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 34                                                             | R   | DI_AlarmCODE15             | Alarm indicator with code No. 15                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 35                                                             | R   | DI_AlarmCODE16             | Alarm indicator with code No. 16                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 36                                                             | R   | DI_AlarmCODE17             | Alarm indicator with code No. 17                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 37                                                             | R   | DI_AlarmCODE18             | Alarm indicator with code No. 18                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 38                                                             | R   | DI_AlarmCODE19             | Alarm indicator with code No. 19                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 39                                                             | R   | DI_AlarmCODE20             | Alarm indicator with code No. 20                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 40                                                             | R   | DI_AlarmCODE21             | Alarm indicator with code No. 21                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 41                                                             | R   | DI_AlarmCODE22             | Alarm indicator with code No. 22                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 42                                                             | R   | DI_AlarmCODE23             | Alarm indicator with code No. 23                                                | 0             | 1             | —             | —                 | Bool  | 1         |
| 43                                                             | R   | DI_AlarmCODE24             | Alarm indicator with code No. 24                                                | 0             | 1             | —             | —                 | Bool  | 1         |

| Address                                                         | R/W | Variable            | Description                                                                                                                                                                     | Minimum value | Maximum value | Pre-set value | Measurement units | Model              | Dimension |
|-----------------------------------------------------------------|-----|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|--------------------|-----------|
| 44                                                              | R   | DI_AlarmCODE25      | Alarm indicator with code No. 25                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 45                                                              | R   | DI_AlarmCODE26      | Alarm indicator with code No. 26                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 46                                                              | R   | DI_AlarmCODE27      | Alarm indicator with code No. 27                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 47                                                              | R   | DI_AlarmCODE28      | Alarm indicator with code No. 28                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 48                                                              | R   | DI_AlarmCODE29      | Alarm indicator with code No. 29                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 49                                                              | R   | DI_AlarmCODE30      | Alarm indicator with code No. 30                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 50                                                              | R   | DI_AlarmCODE31      | Alarm indicator with code No. 31                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 51                                                              | R   | DI_AlarmCODE32      | Alarm indicator with code No. 32                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 52                                                              | R   | DI_AlarmCODE33      | Alarm indicator with code No. 33                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 53                                                              | R   | DI_AlarmCODE34      | Alarm indicator with code No. 34                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 54                                                              | R   | DI_AlarmCODE35      | Alarm indicator with code No. 35                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 55                                                              | R   | DI_AlarmCODE36      | Alarm indicator with code No. 36                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 56                                                              | R   | DI_AlarmCODE37      | Alarm indicator with code No. 37                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 57                                                              | R   | DI_AlarmCODE38      | Alarm indicator with code No. 38                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 58                                                              | R   | DI_AlarmCODE39      | Alarm indicator with code No. 39                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 59                                                              | R   | DI_AlarmCODE40      | Alarm indicator with code No. 40                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 60                                                              | R   | DI_AlarmCODE41      | Alarm indicator with code No. 41                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 61                                                              | R   | DI_AlarmCODE42      | Alarm indicator with code No. 42                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 62                                                              | R   | DI_AlarmCODE43      | Alarm indicator with code No. 43                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 63                                                              | R   | DI_AlarmCODE44      | Alarm indicator with code No. 44                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 64                                                              | R   | DI_AlarmCODE45      | Alarm indicator with code No. 45                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 65                                                              | R   | DI_AlarmCODE46      | Alarm indicator with code No. 46                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 66                                                              | R   | DI_AlarmCODE47      | Alarm indicator with code No. 47                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 67                                                              | R   | DI_AlarmCODE48      | Alarm indicator with code No. 48                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 68                                                              | R   | DI_AlarmCODE49      | Alarm indicator with code No. 49                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 69                                                              | R   | DI_AlarmCODE50      | Alarm indicator with code No. 50                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 70                                                              | R   | DI_AlarmCODE51      | Alarm indicator with code No. 51                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| 71                                                              | R   | DI_AlarmCODE52      | Alarm indicator with code No. 52                                                                                                                                                | 0             | 1             | —             | —                 | Bool               | 1         |
| <b>Input Registers (16-bit registers) - modbus functions: 4</b> |     |                     |                                                                                                                                                                                 |               |               |               |                   |                    |           |
| 0                                                               | R   | IR_CurSelTEMP       | Current temperature of the selected sensor, which controls the air temperature (see HR53).<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                   | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 1                                                               | R   | IR_CurTEMP_SuAirIn  | Current temperature of the main outdoor air sensor before preheating.<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                                        | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 2                                                               | R   | IR_CurTEMP_SuAirOut | Current temperature of the main supply air temperature sensor at the unit outlet downstream of the reheater.<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 3                                                               | R   | IR_CurTEMP_ExAirIn  | Current extract air temperature at the unit inlet.<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                                                           | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 4                                                               | R   | IR_CurTEMP_ExAirOut | Current exhaust air temperature at the unit outlet.<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                                                          | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 5                                                               | R   | IR_CurTEMP_Ext      | Current temperature of the outdoor air temperature sensor (in the control panel, ...).<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                       | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 8                                                               | R   | IR_CurTEMP_Water    | Return heat medium temperature.<br>Value 250 = 25.0 °C. -32768 - no sensor, +32767 - short circuit                                                                              | -32768        | +32767        | —             | °C                | Short Int          | 1         |
| 9                                                               | R   | IR_CurVBAT          | Current battery voltage for RTC.                                                                                                                                                | 0             | 5000          | —             | mV                | Unsigned Short Int | 1         |
| 10                                                              | R   | IR_CurRH_Int        | Current humidity of the main sensor. 0 – no sensor                                                                                                                              | 0             | 100           | —             | %                 | Byte               | 1         |
| 11                                                              | R   | IR_CurRH_Ext        | Current humidity of the outdoor sensor. 0 – no sensor                                                                                                                           | 0             | 100           | —             | %                 | Byte               | 1         |
| 12                                                              | R   | IR_CurCO2_Int       | Current CO <sub>2</sub> level of the main sensor. 0 – no sensor                                                                                                                 | 0             | 10000         | —             | ppm               | Unsigned Short Int | 1         |
| 13                                                              | R   | IR_CurCO2_Ext       | Current CO <sub>2</sub> level of the external sensor. 0 – no sensor                                                                                                             | 0             | 10000         | —             | ppm               | Unsigned Short Int | 1         |
| 14                                                              | R   | IR_CurPM2_5_Int     | Current PM2.5 level of the main sensor. 0 – no sensor                                                                                                                           | 0             | 1000          | —             | µg/m <sup>3</sup> | Unsigned Short Int | 1         |
| 15                                                              | R   | IR_CurPM2_5_Ext     | Current PM2.5 level of the external sensor. 0 – no sensor                                                                                                                       | 0             | 1000          | —             | µg/m <sup>3</sup> | Unsigned Short Int | 1         |
| 16                                                              | R   | IR_CurVOC_Int       | Current VOC level of the main sensor. 0 – no sensor                                                                                                                             | 0             | 100           | —             | %                 | Byte               | 1         |
| 17                                                              | R   | IR_CurVOC_Ext       | Current VOC level of the external sensor. 0 – no sensor                                                                                                                         | 0             | 100           | —             | %                 | Byte               | 1         |
| 18                                                              | R   | IR_Cur10V_SENSOR    | Current value of the 0-10 V sensor                                                                                                                                              | 0             | 100           | —             | %                 | Unsigned Short Int | 1         |
| 19                                                              | R   | IR_CurSuAirFLOW     | Current supply air flow                                                                                                                                                         | 0             | 10000         | —             | m <sup>3</sup> /h | Unsigned Short Int | 1         |
| 20                                                              | R   | IR_CurExAirFLOW     | Current exhaust air flow                                                                                                                                                        | 0             | 10000         | —             | m <sup>3</sup> /h | Unsigned Short Int | 1         |
| 21                                                              | R   | IR_CurSuPRESS       | Current pressure in the supply air duct                                                                                                                                         | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 22                                                              | R   | IR_CurExPRESS       | Current pressure in the exhaust air duct                                                                                                                                        | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |

| Address | R/W | Variable               | Description                                                                                                                                                                                              | Minimum value | Maximum value | Pre-set value | Measurement units | Model              | Dimension |
|---------|-----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|--------------------|-----------|
| 23      | R   | IR_SuRPM               | Supply fan speed                                                                                                                                                                                         | 0             | 5000          | —             | rpm               | Unsigned Short Int | 1         |
| 24      | R   | IR_ExRPM               | Extract fan speed                                                                                                                                                                                        | 0             | 5000          | —             | rpm               | Unsigned Short Int | 1         |
| 25      | R   | IR_CurTIMER_TIME       | Current countdown time of the main timer                                                                                                                                                                 | 0             | 59            | —             | Min.              | Byte               | 2         |
|         |     |                        |                                                                                                                                                                                                          | 0             | 59            | —             | Sec.              | Byte               |           |
|         |     |                        |                                                                                                                                                                                                          | —             |               |               |                   | Byte               |           |
|         |     |                        |                                                                                                                                                                                                          | 0             | 23            | —             | Hours             | Byte               |           |
| 27      | R   | IR_CurFILTER_TIMER     | Countdown time of the filter replacement timer                                                                                                                                                           | 0             | 23            | —             | Hours             | Byte               | 2         |
|         |     |                        |                                                                                                                                                                                                          | 0             | 59            | —             | Min.              | Byte               |           |
|         |     |                        |                                                                                                                                                                                                          | 0             | 365           | —             | Days              | Unsigned Short Int |           |
| 29      | R   | IR_TotalWorkingTime    | Motor hours                                                                                                                                                                                              | 0             | 23            | —             | Hours             | Byte               | 2         |
|         |     |                        |                                                                                                                                                                                                          | 0             | 59            | —             | Min.              | Byte               |           |
|         |     |                        |                                                                                                                                                                                                          | 0             | 65535         | —             | Days              | Unsigned Short Int |           |
| 31      | R   | IR_StateFILTER         | Filter condition:<br>0 - clean, 1 - the intake supply filter is clogged, 2 - the extract filter is clogged, 3 - both filters are clogged or the filter replacement timer has gone off (highest priority) | 0             | 3             | —             | —                 | Byte               | 1         |
| 32      | R   | IR_CurWeekSpeed        | Current speed in Weekly schedule mode:<br>0 - Standby<br>1 - Speed 1<br>2 - Speed 2<br>3 - Speed 3<br>4 - Speed 4<br>5 - Speed 5                                                                         | 0             | 5             | —             | —                 | Byte               | 1         |
| 33      | R   | IR_CurWeekSetTemp      | Current temperature setpoint in Weekly schedule mode:<br>0 - ventilation only, +15 ... + 30 °C                                                                                                           | 0             | 30            | —             | °C                | Byte               | 1         |
| 34      | R   | IR_VerMAIN_FMW         | Firmware version                                                                                                                                                                                         | 0             | 255           | —             | Major             | Byte               | 3         |
|         |     |                        |                                                                                                                                                                                                          | 0             | 255           | —             | Minor             | Byte               |           |
|         |     |                        | Firmware creation date                                                                                                                                                                                   | 1             | 31            | —             | Day               | Byte               |           |
|         |     |                        |                                                                                                                                                                                                          | 1             | 12            | —             | Month             | Byte               |           |
| 37      | R   | IR_DeviceTYPE          | Device type (controller): 1 – S21                                                                                                                                                                        | 0             | 65535         | —             | —                 | Unsigned Short Int | 1         |
| 38      | R   | IR_ALARM               | Alarm/warning indicator:<br>0 – no<br>1 – alarm (highest priority)<br>2 – warning                                                                                                                        | 0             | 2             | —             | —                 | Byte               | 1         |
| 39      | R   | IR_RH_U                | Control signal from the PID humidity controller                                                                                                                                                          | 0             | 100           | —             | %                 | Byte               | 1         |
| 40      | R   | IR_CO2_U               | Control signal from the PID CO <sub>2</sub> level controller                                                                                                                                             | 0             | 100           | —             | %                 | Byte               | 1         |
| 41      | R   | IR_PM2_5_U             | Control signal from the PID PM2.5 level controller                                                                                                                                                       | 0             | 100           | —             | %                 | Byte               | 1         |
| 42      | R   | IR_VOC_U               | Control signal from the PID VOC level controller                                                                                                                                                         | 0             | 100           | —             | %                 | Byte               | 1         |
| 43      | R   | IR_PreHeater_U         | Control signal from the PID preheating controller                                                                                                                                                        | 0             | 100           | —             | %                 | Byte               | 1         |
| 44      | R   | IR_MainHeater_U        | Control signal from the PID reheating controller                                                                                                                                                         | 0             | 100           | —             | %                 | Byte               | 1         |
| 45      | R   | IR_BPS_ROTOR_U         | Control signal from the PID bypass/rotary heat exchanger controller                                                                                                                                      | 0             | 100           | —             | %                 | Byte               | 1         |
| 46      | R   | IR_KKB_U               | Control signal from the PID condenser unit controller                                                                                                                                                    | 0             | 100           | —             | %                 | Byte               | 1         |
| 47      | R   | IR_ReturnWater_U       | Control signal from the PID return heat medium controller                                                                                                                                                | 0             | 100           | —             | %                 | Byte               | 1         |
| 48      | R   | IR_SuAirOutSetTemp     | Temperature setpoint in the supply air duct. Calculated automatically when the room sensor or the sensor in the exhaust air duct is selected.<br>Value 250 = 25.0 °C                                     | 100           | 400           | —             | °C                | Short Int          | 1         |
| 49      | R   | IR_WaterStandbySetTemp | Return heat medium temperature setpoint during winter in Standby mode. Calculated automatically depending on the outdoor temperature.<br>Value 250 = 25.0 °C                                             | 100           | 400           | —             | °C                | Short Int          | 1         |
| 50      | R   | IR_WaterStartSetTemp   | Return heat medium temperature setpoint in winter before the air handling unit start-up. Calculated automatically depending on the outdoor temperature.<br>Value 350 = 35.0 °C                           | 300           | 600           | —             | °C                | Short Int          | 1         |

| Address                                                                 | R/W | Variable            | Description                                                                                                                       | Minimum value | Maximum value | Pre-set value | Measurement units | Model              | Dimension |
|-------------------------------------------------------------------------|-----|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|--------------------|-----------|
| <b>Holding Register (16-bit registers) - modbus functions: 3, 6, 16</b> |     |                     |                                                                                                                                   |               |               |               |                   |                    |           |
| 0                                                                       | R   | HR_VENTILATION_MODE | Ventilation mode:<br>0 - mode 0 ... 100%, 1 - constant flow, 2 - constant pressure                                                | 0             | 2             | 1             | —                 | Byte               | 1         |
| 1                                                                       | R   | HR_MaxSPEED_MODE    | Maximum permissible speed number                                                                                                  | 3             | 5             | 3             | —                 | Byte               | 1         |
| 2                                                                       | R/W | HR_SPEED_MODE       | Speed number:<br>1 – Speed 1, 2 – Speed 2, 3 – Speed 3, 4 - Speed 4, 5 - Speed 5, 255 – manual speed setting mode (see HR17)      | 1             | 255           | 1             | —                 | Byte               | 1         |
| 3                                                                       | R   | HR_MinSPEED         | Minimum possible fan speed                                                                                                        | 0             | 100           | 30            | %                 | Byte               | 1         |
| 4                                                                       | R   | HR_MaxSPEED         | Maximum possible fan speed                                                                                                        | 0             | 100           | 100           | %                 | Byte               | 1         |
| 5                                                                       | R/W | HR_SuSPEED0         | Supply fan speed in Standby mode                                                                                                  | 0             | 100           | 0             | %                 | Byte               | 1         |
| 6                                                                       | R/W | HR_ExSPEED0         | Extract fan speed in Standby mode                                                                                                 | 0             | 100           | 0             | %                 | Byte               | 1         |
| 7                                                                       | R/W | HR_SuSPEED1         | Supply fan speed in Speed 1 mode                                                                                                  | 0             | 100           | 40            | %                 | Byte               | 1         |
| 8                                                                       | R/W | HR_ExSPEED1         | Extract fan speed in Speed 1 mode                                                                                                 | 0             | 100           | 40            | %                 | Byte               | 1         |
| 9                                                                       | R/W | HR_SuSPEED2         | Supply fan speed in Speed 2 mode                                                                                                  | 0             | 100           | 70            | %                 | Byte               | 1         |
| 10                                                                      | R/W | HR_ExSPEED2         | Extract fan speed in Speed 2 mode                                                                                                 | 0             | 100           | 70            | %                 | Byte               | 1         |
| 11                                                                      | R/W | HR_SuSPEED3         | Supply fan speed in Speed 3 mode                                                                                                  | 0             | 100           | 100           | %                 | Byte               | 1         |
| 12                                                                      | R/W | HR_ExSPEED3         | Extract fan speed in Speed 3 mode                                                                                                 | 0             | 100           | 100           | %                 | Byte               | 1         |
| 13                                                                      | R/W | HR_SuSPEED4         | Supply fan speed in Speed 4 mode                                                                                                  | 0             | 100           | 100           | %                 | Byte               | 1         |
| 14                                                                      | R/W | HR_ExSPEED4         | Extract fan speed in Speed 4 mode                                                                                                 | 0             | 100           | 100           | %                 | Byte               | 1         |
| 15                                                                      | R/W | HR_SuSPEED5         | Supply fan speed in Speed 5 mode                                                                                                  | 0             | 100           | 100           | %                 | Byte               | 1         |
| 16                                                                      | R/W | HR_ExSPEED5         | Extract fan speed in Speed 5 mode                                                                                                 | 0             | 100           | 100           | %                 | Byte               | 1         |
| 17                                                                      | R/W | HR_ManualSPEED      | Fan speed in manual speed setting mode<br>The balance between supply and exhaust air corresponds to the current preset speeds 1-5 | 0             | 100           | 50            | %                 | Byte               | 1         |
| 18                                                                      | R/W | HR_BlowingSPEED     | Fan speed while blowing electric heaters                                                                                          | 0             | 100           | 50            | %                 | Byte               | 1         |
| 19                                                                      | R/W | HR_Boost_SuSPEED    | Supply fan speed in Boost mode                                                                                                    | 0             | 100           | 100           | %                 | Byte               | 1         |
| 20                                                                      | R/W | HR_Boost_ExSPEED    | Extract fan speed in Boost mode                                                                                                   | 0             | 100           | 100           | %                 | Byte               | 1         |
| 21                                                                      | R/W | HR_FPLC_SuSPEED     | Supply fan speed in Fireplace mode                                                                                                | 0             | 100           | 60            | %                 | Byte               | 1         |
| 22                                                                      | R/W | HR_FPLC_ExSPEED     | Extract fan speed in Fireplace mode                                                                                               | 0             | 100           | 40            | %                 | Byte               | 1         |
| 23                                                                      | R   | HR_MinAirFLOW       | Minimum possible air flow of the unit                                                                                             | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 24                                                                      | R   | HR_MaxAirFLOW       | Maximum possible air flow of the unit                                                                                             | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 25                                                                      | R/W | HR_SuSPEED0_FLOW    | Supply air flow in Standby mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 26                                                                      | R/W | HR_ExSPEED0_FLOW    | Extract air flow in Standby mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 27                                                                      | R/W | HR_SuSPEED1_FLOW    | Supply air flow in Speed 1 mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 28                                                                      | R/W | HR_ExSPEED1_FLOW    | Extract air flow in Speed 1 mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 29                                                                      | R/W | HR_SuSPEED2_FLOW    | Supply air flow in Speed 2 mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 30                                                                      | R/W | HR_ExSPEED2_FLOW    | Extract air flow in Speed 2 mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 31                                                                      | R/W | HR_SuSPEED3_FLOW    | Supply air flow in Speed 3 mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 32                                                                      | R/W | HR_ExSPEED3_FLOW    | Extract air flow in Speed 3 mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 33                                                                      | R/W | HR_SuSPEED4_FLOW    | Supply air flow in Speed 4 mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 34                                                                      | R/W | HR_ExSPEED4_FLOW    | Extract air flow in Speed 4 mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 35                                                                      | R/W | HR_SuSPEED5_FLOW    | Supply air flow in Speed 5 mode                                                                                                   | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 36                                                                      | R/W | HR_ExSPEED5_FLOW    | Extract air flow in Speed 5 mode                                                                                                  | 0             | 10000         | —             | m³/h              | Unsigned Short Int | 1         |
| 37                                                                      | R   | HR_MinAirPRESS      | Minimum possible pressure in the air duct                                                                                         | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 38                                                                      | R   | HR_MaxAirPRESS      | Maximum possible pressure in the air duct                                                                                         | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 39                                                                      | R/W | HR_SuSPEED0_PRESS   | Pressure in the supply air duct in Standby mode                                                                                   | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 40                                                                      | R/W | HR_ExSPEED0_PRESS   | Pressure in the exhaust air duct in Standby mode                                                                                  | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 41                                                                      | R/W | HR_SuSPEED1_PRESS   | Pressure in the supply air duct in Speed 1 mode                                                                                   | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |
| 42                                                                      | R/W | HR_ExSPEED1_PRESS   | Pressure in the exhaust air duct in Speed 1 mode                                                                                  | 0             | 10000         | —             | Pa                | Unsigned Short Int | 1         |

| Address | R/W | Variable                  | Description                                                                                                                                                                                                                                                         | Minimum value    | Maximum value       | Pre-set value    | Measurement units                | Model                        | Dimension |
|---------|-----|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|------------------|----------------------------------|------------------------------|-----------|
| 43      | R/W | HR_OPERATION_MODE         | Unit operation mode:<br>0 - ventilation only 1 - heating, 2 - cooling, 3 - auto                                                                                                                                                                                     | 0                | 3                   | 3                | —                                | Byte                         | 1         |
| 44      | R/W | HR_SetTEMP                | Room temperature setpoint in normal mode                                                                                                                                                                                                                            | 15               | 30                  | 23               | °C                               | Byte                         | 1         |
| 45      | R/W | HR_SetRH                  | Humidity threshold setpoint                                                                                                                                                                                                                                         | 40               | 80                  | 60               | %RH                              | Byte                         | 1         |
| 46      | R/W | HR_SetCO2                 | CO <sub>2</sub> threshold setpoint                                                                                                                                                                                                                                  | 400              | 2000                | 1200             | ppm                              | Byte                         | 1         |
| 47      | R/W | HR_SetPM2_5               | PM2.5 threshold setpoint                                                                                                                                                                                                                                            | 100              | 1000                | 400              | µg/m <sup>3</sup>                | Byte                         | 1         |
| 48      | R/W | HR_SetVOC                 | VOC threshold setpoint                                                                                                                                                                                                                                              | 20               | 100                 | 40               | %                                | Byte                         | 1         |
| 49      | R/W | HR_TIMER_MODE             | Timer mode:<br>0 - Standby, 1 - Speed 1, 2 - Speed 2, 3 - Speed 3, 4 - Speed 4, 5 - Speed 5                                                                                                                                                                         | 0                | 5                   | 1                | —                                | Byte                         | 1         |
| 50      | R/W | HR_SetTIMER_TEMP          | Room temperature setpoint for the main timer:<br>0 - ventilation only, +15...+ 30 °C                                                                                                                                                                                | 0                | 30                  | 23               | °C                               | Byte                         | 1         |
| 51      | R/W | HR_SetTIMER_TIME          | Time setpoint of the main timer                                                                                                                                                                                                                                     | 0<br>0           | 23<br>59            | 0<br>30          | Hours<br>Min.                    | Byte<br>Byte                 | 1         |
| 52      | R/W | HR_SetTEMP_WinterSummer   | Transition temperature winter/summer                                                                                                                                                                                                                                | 5                | 15                  | 7                | °C                               | Byte                         | 1         |
| 53      | R/W | HR_SetTEMP_SENSOR         | Selecting a temperature sensor for controlling room temperature:<br>0 - in the exhaust air duct, 1 - external sensor in the control panel, 2 - in the supply air duct                                                                                               | 0                | 2                   | 2                | —                                | Byte                         | 1         |
| 54      | R/W | HR_MainHEATER_TYPE        | Main heater type:<br>0 - turn off, 1 - electric, 2 - water                                                                                                                                                                                                          | 0                | 2                   | —                | —                                | Byte                         | 1         |
| 55      | R/W | HR_COOLER_TYPE            | Cooler control type:<br>0 - turn off, 1 - discrete, 2 - analogue 0-10 V (integrated)                                                                                                                                                                                | 0                | 2                   | —                | —                                | Byte                         | 1         |
| 56      | R/W | HR_DEF_MODE               | Heat exchanger Freeze protection mode:<br>0 - turn off, 1 - preheating, 2 - bypass/rotor, 3 - fan imbalance                                                                                                                                                         | 0                | 3                   | —                | —                                | Byte                         | 1         |
| 57      | R   | HR_BPS_ROTOR_TYPE         | Bypass/rotary heat exchanger type:<br>0 - not available, 1 - bypass with two-point control, 2 - bypass with analogue control, 3 - rotary heat exchanger with discrete control, 4 - rotary heat exchanger with analogue control, 5 - bypass with three-point control | 0                | 4                   | —                | —                                | Byte                         | 1         |
| 58      | R/W | HR_SetFILTER_TIMER        | Filter timer setpoint: 0 - turn off the timer, 70...365 days                                                                                                                                                                                                        | 0                | 365                 | 90               | Days                             | Unsigned Short Int           | 1         |
| 59      | R/W | HR_BoostDelaySwitchingOff | Setpoint of the Boost mode turn-off delay                                                                                                                                                                                                                           | 0                | 60                  | 0                | Min.                             | Byte                         | 1         |
| 60      | R/W | HR_BoostDelaySwitchingOn  | Setpoint of the Boost mode turn-on delay                                                                                                                                                                                                                            | 0                | 15                  | 0                | Min.                             | Byte                         | 1         |
| 61      | R/W | HR_RTC_TIME               | RTC time                                                                                                                                                                                                                                                            | 0<br>0<br>—<br>0 | 59<br>59<br>—<br>23 | —<br>—<br>—<br>— | Min.<br>Min.<br>Sec.<br>Hours    | Byte<br>Byte<br>Byte<br>Byte | 2         |
| 63      | R/W | HR_RTC_CALENDAR           | RTC calendar                                                                                                                                                                                                                                                        | 1<br>1<br>1<br>0 | 31<br>7<br>12<br>99 | —<br>—<br>—<br>— | Day<br>Week day<br>Month<br>Year | Byte<br>Byte<br>Byte<br>Byte | 2         |
| 65      | R/W | HR_MaxCO2_Int             | Maximum value of the main CO <sub>2</sub> sensor                                                                                                                                                                                                                    | 500              | 10000               | 2000             | ppm                              | Unsigned Short Int           | 1         |
| 66      | R/W | HR_MaxPM2_5_Int           | Maximum value of the main PM2.5 sensor                                                                                                                                                                                                                              | 500              | 10000               | 1000             | µg/m <sup>3</sup>                | Unsigned Short Int           | 1         |
| 67      | R/W | HR_SetMinSuAirOutTEMP     | Minimum room supply air temperature control setpoint                                                                                                                                                                                                                | 5                | 12                  | 10               | °C                               | Byte                         | 1         |
| 68      | R/W | HR_MainHeaterMODE         | Main heater operation mode:<br>1 - control 0 - 100 %, 2 - AUTO                                                                                                                                                                                                      | 1                | 2                   | 2                | —                                | Byte                         | 1         |
| 69      | R/W | HR_SetMainHeaterMANUAL    | Manual control of the main heater                                                                                                                                                                                                                                   | 0                | 100                 | 50               | %                                | Byte                         | 1         |
| 70      | R/W | HR_CoolerMODE             | Cooler operation mode:<br>1 - turn on the cooler with discrete configuration, control 0-100 % with analogue configuration, 2 - AUTO                                                                                                                                 | 1                | 2                   | 2                | —                                | Byte                         | 1         |
| 71      | R/W | HR_SetCoolerMANUAL        | Manual cooler control with analogue configuration                                                                                                                                                                                                                   | 0                | 100                 | 0                | %                                | Byte                         | 1         |
| 72      | R/W | HR_PreHeaterMODE          | Preheating operation mode:<br>1 - control 0 - 100 %, 2 - AUTO                                                                                                                                                                                                       | 1                | 2                   | 2                | —                                | Byte                         | 1         |
| 73      | R/W | HR_SetPreHeaterMANUAL     | Manual preheating control                                                                                                                                                                                                                                           | 0                | 100                 | 50               | %                                | Byte                         | 1         |
| 74      | R/W | HR_BPS_ROTOR_MODE         | Bypass/rotary heat exchanger operation mode:<br>0 - close the bypass/start the rotor, 1 - open the bypass/stop the rotor with discrete configuration, control 0-100 % with analogue configuration, 2 - AUTO                                                         | 0                | 2                   | 2                | —                                | Byte                         | 1         |
| 75      | R/W | HR_SetBpsRotorMANUAL      | Manual bypass/rotor control with analogue configuration:<br>0 % - bypass closed/rotor rotates at maximum speed, 100 %/bypass open, rotor stopped                                                                                                                    | 0                | 100                 | 100              | %                                | Byte                         | 1         |
| 76      | R/W | HR_RH_Kp                  | Kp coefficient of the PID humidity controller                                                                                                                                                                                                                       | 0                | 1000                | 150              | —                                | Unsigned Short Int           | 1         |
| 77      | R/W | HR_RH_Ki                  | Ki coefficient of the PID humidity controller                                                                                                                                                                                                                       | 0                | 1000                | 150              | —                                | Unsigned Short Int           | 1         |
| 78      | R/W | HR_RH_Kd                  | Kd coefficient of the PID humidity controller                                                                                                                                                                                                                       | 0                | 1000                | 0                | —                                | Unsigned Short Int           | 1         |
| 79      | R/W | HR_CO2_Kp                 | Kp coefficient of the PID CO <sub>2</sub> level controller                                                                                                                                                                                                          | 0                | 1000                | 150              | —                                | Unsigned Short Int           | 1         |

| Address | R/W | Variable                 | Description                                                                                                                                                     | Minimum value | Maximum value | Pre-set value | Measurement units | Model              | Dimension |
|---------|-----|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|-------------------|--------------------|-----------|
| 80      | R/W | HR_CO2_Ki                | Ki coefficient of the PID CO <sub>2</sub> level controller                                                                                                      | 0             | 1000          | 150           | —                 | Unsigned Short Int | 1         |
| 81      | R/W | HR_CO2_Kd                | Kd coefficient of the PID CO <sub>2</sub> level controller                                                                                                      | 0             | 1000          | 0             | —                 | Unsigned Short Int | 1         |
| 82      | R/W | HR_PM2_5_Kp              | Kp coefficient of the PID PM2.5 level controller                                                                                                                | 0             | 1000          | 150           | —                 | Unsigned Short Int | 1         |
| 83      | R/W | HR_PM2_5_Ki              | Ki coefficient of the PID PM2.5 level controller                                                                                                                | 0             | 1000          | 150           | —                 | Unsigned Short Int | 1         |
| 84      | R/W | HR_PM2_5_Kd              | Kd coefficient of the PID PM2.5 level controller                                                                                                                | 0             | 1000          | 0             | —                 | Unsigned Short Int | 1         |
| 85      | R/W | HR_VOC_Kp                | Kp coefficient of the PID VOC level controller                                                                                                                  | 0             | 1000          | 150           | —                 | Unsigned Short Int | 1         |
| 86      | R/W | HR_VOC_Ki                | Ki coefficient of the PID VOC level controller                                                                                                                  | 0             | 1000          | 150           | —                 | Unsigned Short Int | 1         |
| 87      | R/W | HR_VOC_Kd                | Kd coefficient of the PID VOC level controller                                                                                                                  | 0             | 1000          | 0             | —                 | Unsigned Short Int | 1         |
| 88      | R/W | HR_PreHeater_Kp          | Kp coefficient of the PID preheating controller                                                                                                                 | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 89      | R/W | HR_PreHeater_Ki          | Ki coefficient of the PID preheating controller                                                                                                                 | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 90      | R/W | HR_PreHeater_Kd          | Kd coefficient of the PID preheating controller                                                                                                                 | 0             | 1000          | 500           | —                 | Unsigned Short Int | 1         |
| 91      | R/W | HR_MainHeater_Kp         | Kp coefficient of the PID reheating controller                                                                                                                  | 0             | 1000          | 400           | —                 | Unsigned Short Int | 1         |
| 92      | R/W | HR_MainHeater_Ki         | Ki coefficient of the PID reheating controller                                                                                                                  | 0             | 1000          | 400           | —                 | Unsigned Short Int | 1         |
| 93      | R/W | HR_MainHeater_Kd         | Kd coefficient of the PID reheating controller                                                                                                                  | 0             | 1000          | 600           | —                 | Unsigned Short Int | 1         |
| 94      | R/W | HR_BPS_ROTOR_Kp          | Kp coefficient of the PID bypass/rotary heat exchanger controller                                                                                               | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 95      | R/W | HR_BPS_ROTOR_Ki          | Ki coefficient of the PID bypass/rotary heat exchanger controller                                                                                               | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 96      | R/W | HR_BPS_ROTOR_Kd          | Kd coefficient of the PID bypass/rotary heat exchanger controller                                                                                               | 0             | 1000          | 500           | —                 | Unsigned Short Int | 1         |
| 97      | R/W | HR_KKB_Kp                | Kp coefficient of the PID condenser unit controller                                                                                                             | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 98      | R/W | HR_KKB_Ki                | Ki coefficient of the PID condenser unit controller                                                                                                             | 0             | 1000          | 200           | —                 | Unsigned Short Int | 1         |
| 99      | R/W | HR_KKB_Kd                | Kd coefficient of the PID condenser unit controller                                                                                                             | 0             | 1000          | 500           | —                 | Unsigned Short Int | 1         |
| 100     | R/W | HR_ReturnWater_Kp        | Kp coefficient of the PID return heat medium controller                                                                                                         | 0             | 1000          | 120           | —                 | Unsigned Short Int | 1         |
| 101     | R/W | HR_ReturnWater_Ki        | Ki coefficient of the PID return heat medium controller                                                                                                         | 0             | 1000          | 120           | —                 | Unsigned Short Int | 1         |
| 102     | R/W | HR_ReturnWater_Kd        | Kd coefficient of the PID return heat medium controller                                                                                                         | 0             | 1000          | 350           | —                 | Unsigned Short Int | 1         |
| 103     | R   | HR_FanAlarmCTRL          | Fan alarm control type:<br>0 - no alarm control, 1...254 - number of tach pulses per fan rotation, 255 - fan alarm control using a differential pressure switch | 0             | 255           | 2             | —                 | Byte               | 1         |
| 104     | R   | HR_SetTimeDetectFanALARM | Time for fan alarm detection                                                                                                                                    | 5             | 120           | 30            | Sec.              | Byte               | 1         |
| 105     | R/W | HR_SetTimeOpenVALVE      | Damper opening time (fan turn-on delay)                                                                                                                         | 0             | 240           | 0             | Sec.              | Byte               | 1         |
| 106     | R/W | HR_SetTimeFanBLOWING     | Electric heater blowdown time                                                                                                                                   | 20            | 240           | 120           | Sec.              | Byte               | 1         |
| 107     | R/W | HR_KKB_MinTimeOFF        | Minimum downtime of the condenser unit before restarting                                                                                                        | 0             | 20            | 3             | Min.              | Byte               | 1         |
| 108     | R/W | HR_KKB_MinTimeON         | Minimum operating time of the condenser unit before shutdown                                                                                                    | 0             | 20            | 1             | Min.              | Byte               | 1         |
| 109     | R/W | HR_KKB_HYSTERESIS        | Hysteresis for turning the condenser unit on/off with discrete control                                                                                          | 1             | 10            | 2             | °C                | Byte               | 1         |
| 110     | R   | HR_BPS_Position          | Bypass location: 0 - from outdoors, 1 - from indoors                                                                                                            | 0             | 1             | —             | —                 | Byte               | 1         |
| 111     | R   | HR_TimeOpenBPS           | Opening time of the bypass with three-point control                                                                                                             | 2             | 300           | —             | Sec.              | Byte               | 1         |
| 112     | R/W | HR_CorrTEMP_SuAirIn      | Correction of the intake air temperature sensor at the unit inlet. Value 250 = 25.0 °C                                                                          | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 113     | R/W | HR_CorrTEMP_SuAirOut     | Correction of the supply air temperature sensor at the unit outlet (downstream of the heat exchanger/downstream of the heater). Value 250 = 25.0 °C             | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 114     | R/W | HR_CorrTEMP_ExAirIn      | Correction of the extract air temperature sensor at the unit inlet. Value 250 = 25.0 °C                                                                         | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 115     | R/W | HR_CorrTEMP_ExAirOut     | Correction of the exhaust air temperature sensor at the unit outlet. Value 250 = 25.0 °C                                                                        | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 116     | R/W | HR_CorrTEMP_Water        | Correction of the return heat medium temperature sensor. Value 250 = 25.0 °C                                                                                    | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 117     | R/W | HR_CorrTEMP_Ext          | Correction of the outdoor air temperature sensor Value 250 = 25.0 °C                                                                                            | -500          | +500          | 0             | °C                | Short Int          | 1         |
| 118     | R/W | HR_WaterValveMinPos      | Minimum position of the water heater valve in winter                                                                                                            | 0             | 100           | 0             | %                 | Byte               | 1         |

| Address | R/W | Variable             | Description                                                                                                                                                   | Minimum value        | Maximum value        | Pre-set value        | Measurement units            | Model        | Dimension |
|---------|-----|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|----------------------|------------------------------|--------------|-----------|
| 119     | R/W | HR_WaterMaxStartTime | Time for detecting return heat medium underheating alarm before the AHU start in winter                                                                       | 2                    | 30                   | 5                    | Min.                         | Byte         | 1         |
| 120     | R/W | HR_WaterMinStartTemp | Initial value of the return heat medium temperature required for the AHU start in winter at outdoor temperature $\geq +10^{\circ}\text{C}$                    | 30                   | 60                   | 30                   | $^{\circ}\text{C}$           | Short Int    | 1         |
| 121     | R/W | HR_WaterMaxStartTemp | Final value of the return heat medium temperature required for the AHU start in winter at outdoor temperature $\leq -30^{\circ}\text{C}$                      | 30                   | 60                   | 50                   | $^{\circ}\text{C}$           | Short Int    | 1         |
| 122     | R/W | HR_WaterMinAlarmTemp | Initial value of the return heat medium temperature for the AHU shutdown caused by a freeze alarm in winter at outdoor temperature $\geq +10^{\circ}\text{C}$ | 10                   | 30                   | 12                   | $^{\circ}\text{C}$           | Short Int    | 1         |
| 123     | R/W | HR_WaterMaxAlarmTemp | Final value of the return heat medium temperature for the AHU shutdown caused by a freeze alarm in winter at outdoor temperature $\leq -30^{\circ}\text{C}$   | 10                   | 30                   | 20                   | $^{\circ}\text{C}$           | Short Int    | 1         |
| 124     | R/W | HR_ENGINEER_PWD      | Password to enter the engineering menu. The string should be 1-4 characters long. The end of the string is determined by the Null character                   | 48<br>48<br>48<br>48 | 57<br>57<br>57<br>57 | 49<br>49<br>49<br>49 | Char<br>Char<br>Char<br>Char | String       | 2         |
| 126     | R/W | HR_SetWEEK_Mo        | Speed number for Mo. in the 1st time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Mo. in the 1st period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 127     | R/W |                      | Hours of the end of the 1st period on Mo.                                                                                                                     | 0                    | 23                   | 6                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 1st period on Mo.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 128     | R/W |                      | Speed number for Mo. in the 2nd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Mo. in the 2nd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 129     | R/W |                      | Hours of the end of the 2nd period on Mo.                                                                                                                     | 0                    | 23                   | 9                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 2nd period on Mo.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 130     | R/W |                      | Speed number for Mo. in the 3rd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Mo. in the 3rd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 131     | R/W |                      | Hours of the end of the 3rd period on Mo.                                                                                                                     | 0                    | 23                   | 19                   | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 3rd period on Mo.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 132     | R/W |                      | Speed number for Mo. in the 4th time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Mo. in the 4th period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 133     | R   |                      | Reserved. The end of the 4th period is always at 23:59                                                                                                        | 0<br>0               | 23<br>59             | 23<br>59             | Hours<br>Min.                | Byte<br>Byte | 1         |
| 134     | R/W | HR_SetWEEK_Tu        | Speed number for Tu. in the 1st time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Tu. in the 1st period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 135     | R/W |                      | Hours of the end of the 1st period on Tu.                                                                                                                     | 0                    | 23                   | 6                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 1st period on Tu.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 136     | R/W |                      | Speed number for Tu. in the 2nd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Tu. in the 2nd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 137     | R/W |                      | Hours of the end of the 2nd period on Tu.                                                                                                                     | 0                    | 23                   | 9                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 2nd period on Tu.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 138     | R/W |                      | Speed number for Tu. in the 3rd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Tu. in the 3rd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 139     | R/W |                      | Hours of the end of the 3rd period on Tu.                                                                                                                     | 0                    | 23                   | 19                   | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 3rd period on Tu.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 140     | R/W |                      | Speed number for Tu. in the 4th time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for Tu. in the 4th period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 141     | R   |                      | Reserved. The end of the 4th period is always at 23:59                                                                                                        | 0<br>0               | 23<br>59             | 23<br>59             | Hours<br>Min.                | Byte<br>Byte | 1         |
| 142     | R/W | HR_SetWEEK_We        | Speed number for We. in the 1st time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for We. in the 1st period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 143     | R/W |                      | Hours of the end of the 1st period on We. 0 23 6 Hours Byte 1                                                                                                 | 0                    | 23                   | 6                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 1st period on We.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 144     | R/W |                      | Speed number for We. in the 2nd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for We. in the 2nd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 145     | R/W |                      | Hours of the end of the 2nd period on We.                                                                                                                     | 0                    | 23                   | 9                    | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 2nd period on We.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 146     | R/W |                      | Speed number for We. in the 3rd time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for We. in the 3rd period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 147     | R/W |                      | Hours of the end of the 3rd period on We.                                                                                                                     | 0                    | 23                   | 19                   | Hours                        | Byte         | 1         |
|         |     |                      | Minutes of the end of the 3rd period on We.                                                                                                                   | 0                    | 59                   | 0                    | Min.                         | Byte         |           |
| 148     | R/W |                      | Speed number for We. in the 4th time period                                                                                                                   | 0                    | 5                    | 1                    | —                            | Byte         | 1         |
|         |     |                      | Temperature setpoint for We. in the 4th period                                                                                                                | 15                   | 30                   | 23                   | $^{\circ}\text{C}$           | Byte         |           |
| 149     | R   |                      | Reserved. The end of the 4th period is always at 23:59                                                                                                        | 0<br>0               | 23<br>59             | 23<br>59             | Hours<br>Min.                | Byte<br>Byte | 1         |

| Address | R/W | Variable      | Description                                            | Minimum value | Maximum value | Pre-set value | Measurement units | Model | Dimension |
|---------|-----|---------------|--------------------------------------------------------|---------------|---------------|---------------|-------------------|-------|-----------|
| 150     | R/W | HR_SetWEEK_Th | Speed number for Th. in the 1st time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Th. in the 1st period         | 0             | 30            | 23            | °C                | Byte  |           |
| 151     | R/W |               | Hours of the end of the 1st period on Th.              | 0             | 23            | 6             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 1st period on Th.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 152     | R/W |               | Speed number for Th. in the 2nd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Th. in the 2nd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 153     | R/W |               | Hours of the end of the 2nd period on Th.              | 0             | 23            | 9             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 2nd period on Th.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 154     | R/W |               | Speed number for Th. in the 3rd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Th. in the 3rd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 155     | R/W |               | Hours of the end of the 3rd period on Th.              | 0             | 23            | 19            | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 3rd period on Th.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 156     | R/W | HR_SetWEEK_Fr | Speed number for Th. in the 4th time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Th. in the 4th period         | 0             | 30            | 23            | °C                | Byte  |           |
| 157     | R   |               | Reserved. The end of the 4th period is always at 23:59 | 0             | 23            | 23            | Hours             | Byte  |           |
|         |     |               |                                                        | 0             | 59            | 59            | Min.              | Byte  |           |
| 158     | R/W | HR_SetWEEK_Fr | Speed number for Fr. in the 1st time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Fr. in the 1st period         | 0             | 30            | 23            | °C                | Byte  |           |
| 159     | R/W |               | Hours of the end of the 1st period on Fr.              | 0             | 23            | 6             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 1st period on Fr.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 160     | R/W |               | Speed number for Fr. in the 2nd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Fr. in the 2nd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 161     | R/W |               | Hours of the end of the 2nd period on Fr.              | 0             | 23            | 9             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 2nd period on Fr.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 162     | R/W |               | Speed number for Fr. in the 3rd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Fr. in the 3rd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 163     | R/W |               | Hours of the end of the 3rd period on Fr.              | 0             | 23            | 19            | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 3rd period on Fr.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 164     | R/W | HR_SetWEEK_Sa | Speed number for Fr. in the 4th time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Fr. in the 4th period         | 0             | 30            | 23            | °C                | Byte  |           |
| 165     | R   |               | Reserved. The end of the 4th period is always at 23:59 | 0             | 23            | 23            | Hours             | Byte  |           |
|         |     |               |                                                        | 0             | 59            | 59            | Min.              | Byte  |           |
| 166     | R/W | HR_SetWEEK_Sa | Speed number for Sa. in the 1st time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Sa. in the 1st period         | 0             | 30            | 23            | °C                | Byte  |           |
| 167     | R/W |               | Hours of the end of the 1st period on Sa.              | 0             | 23            | 6             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 1st period on Sa.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 168     | R/W |               | Speed number for Sa. in the 2nd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Sa. in the 2nd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 169     | R/W |               | Hours of the end of the 2nd period on Sa.              | 0             | 23            | 9             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 2nd period on Sa.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 170     | R/W |               | Speed number for Sa. in the 3rd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Sa. in the 3rd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 171     | R/W |               | Hours of the end of the 3rd period on Sa.              | 0             | 23            | 19            | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 3rd period on Sa.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 172     | R/W | HR_SetWEEK_Su | Speed number for Sa. in the 4th time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Sa. in the 4th period         | 0             | 30            | 23            | °C                | Byte  |           |
| 173     | R   |               | Reserved. The end of the 4th period is always at 23:59 | 0             | 23            | 23            | Hours             | Byte  |           |
|         |     |               |                                                        | 0             | 59            | 59            | Min.              | Byte  |           |
| 174     | R/W | HR_SetWEEK_Su | Speed number for Su. in the 1st time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Su. in the 1st period         | 0             | 30            | 23            | °C                | Byte  |           |
| 175     | R/W |               | Hours of the end of the 1st period on Su.              | 0             | 23            | 6             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 1st period on Su.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 176     | R/W |               | Speed number for Su. in the 2nd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Su. in the 2nd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 177     | R/W |               | Hours of the end of the 2nd period on Su.              | 0             | 23            | 9             | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 2nd period on Su.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 178     | R/W |               | Speed number for Su. in the 3rd time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Su. in the 3rd period         | 0             | 30            | 23            | °C                | Byte  |           |
| 179     | R/W |               | Hours of the end of the 3rd period on Su.              | 0             | 23            | 19            | Hours             | Byte  |           |
|         |     |               | Minutes of the end of the 3rd period on Su.            | 0             | 59            | 0             | Min.              | Byte  |           |
| 180     | R/W | HR_SetWEEK_Su | Speed number for Su. in the 4th time period            | 0             | 5             | 1             | —                 | Byte  | 1         |
|         |     |               | Temperature setpoint for Su. in the 4th period         | 0             | 30            | 23            | °C                | Byte  |           |
| 181     | R   |               | Reserved. The end of the 4th period is always at 23:59 | 0             | 23            | 23            | Hours             | Byte  |           |
|         |     |               |                                                        | 0             | 59            | 59            | Min.              | Byte  |           |



